

PharmaTrace AI

Exploratory Data Analysis Report

Phase 4 Production Master Dataset
18 Tables | 52,548 Rows | 3,000 Products

Capstone Project
August 2026

Executive Summary

This report presents a comprehensive exploratory data analysis of the PharmaTrace AI master dataset. The dataset models a complete pharmaceutical supply chain across 18 interconnected tables totaling 52,548 rows, built around 3,000 real FDA-registered products.

Here are the key takeaways from our analysis:

Product Catalog: We have 3,000 products with real FDA identities. Oral tablets make up about 68% of dosage forms. Prices range from \$5 to \$500, averaging around \$78.

Manufacturing: 6,000 manufacturing orders show a consistent 98.5% yield. All finished batches pass QC. About 20% carry recall flags.

Supply Chain: 12 GMP-certified suppliers provide 799 raw materials through 1,598 batches. Quality ratings are solid at 3.5-4.5 out of 5.

Distribution: 2,100 shipments move products from 8 warehouses through 25 distributors to 300 retail endpoints - hospitals, clinics, and pharmacies.

Recalls: 600 real FDA recall records from openFDA. Class II dominates. CGMP deviations and impurity contamination (especially NDMA) are the top causes.

Returns and Disposal: 532 returns total - 79% triggered by recalls, 21% by expiry. Each return goes through disposal with a destruction certificate.

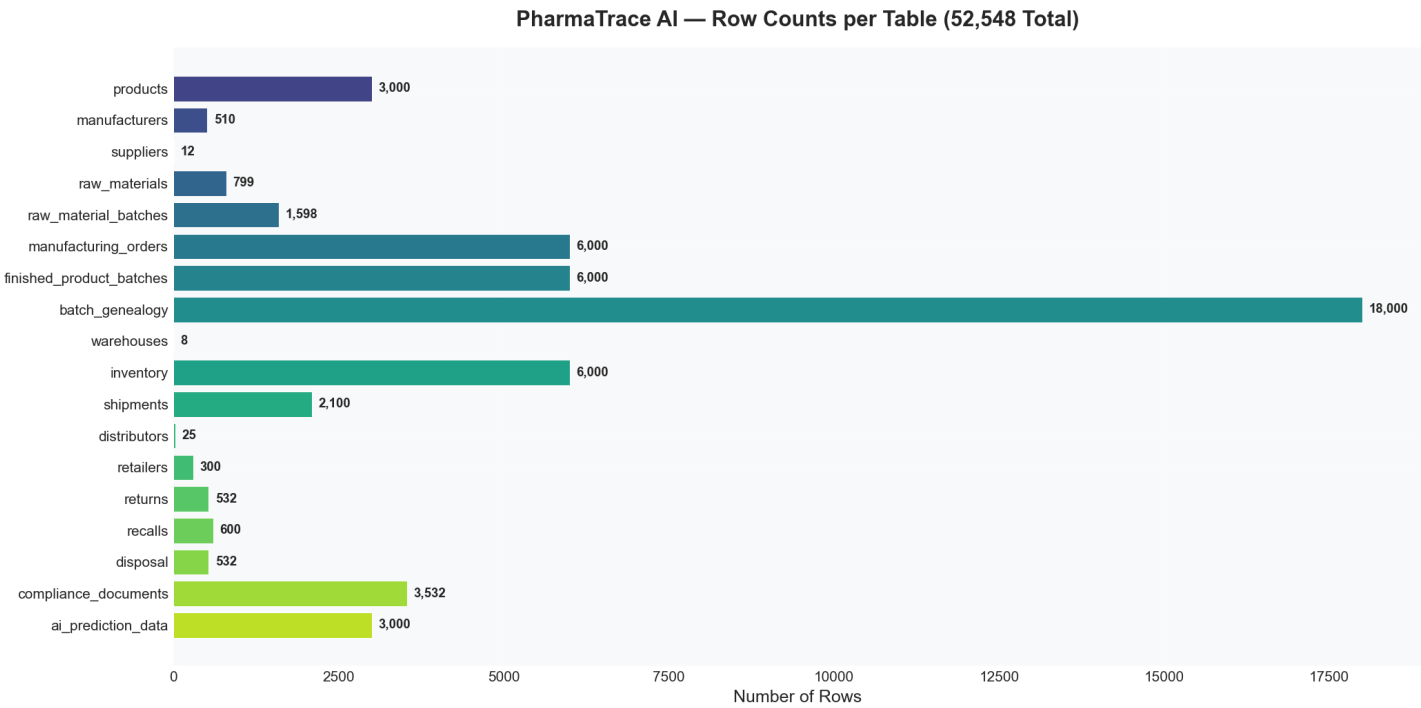
AI Readiness: 3,000 prediction records with risk scores. 206 contain real demand data. The dataset supports demand forecasting, expiry risk scoring, recall prediction, and supplier risk assessment.

Data Quality: Zero duplicate keys, zero null primary keys, zero broken foreign keys across all 31 relationships. The dataset is clean and integration-ready.

Dataset Overview

Table	Rows	Cols	What It Contains
products	3,000	21	Real FDA product identities (NDC, UNII, RxCUI)
manufacturers	510	8	Real labelers and Drugs@FDA sponsors
suppliers	12	8	API and excipient vendors with quality ratings
raw_materials	799	7	Materials with real UNII codes
raw_material_batches	1,598	9	QC-tracked incoming material batches
manufacturing_orders	6,000	9	Production work orders with yield data
finished_product_batches	6,000	11	GS1-serialized batches with recall flags
batch_genealogy	18,000	5	Raw material to finished product trace links
warehouses	8	9	Temperature-controlled storage facilities
inventory	6,000	9	Stock levels with 432 real seed records
shipments	2,100	11	Warehouse-to-retailer shipments
distributors	25	8	Regional wholesale distributors
retailers	300	9	Hospitals, clinics, pharmacies
returns	532	12	420 recall-driven + 112 expiry-driven
recalls	600	12	Real FDA Enforcement records
disposal	532	11	Destruction records with certificates
compliance_documents	3,532	9	1,148 real FDA doc URLs
ai_prediction_data	3,000	12	Risk scores + 206 real demand points
TOTAL	52,548		

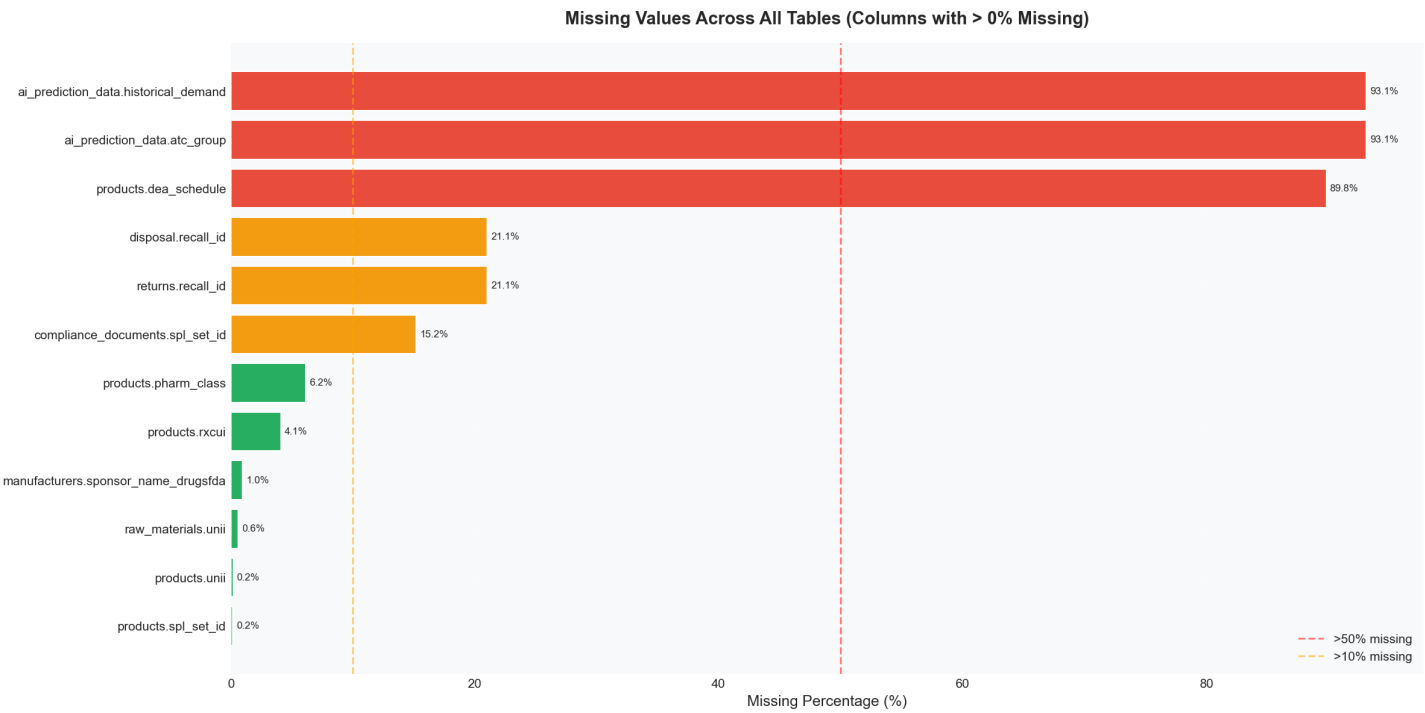
Dataset Overview - Row Counts per Table



Key Insights:

Looking at the overall dataset structure, batch_genealogy stands out as the largest table with 18,000 rows since each finished batch links to 3 raw material batches. The core operational tables (products, manufacturing orders, finished batches, inventory) each contain around 6,000 rows, which gives us a solid amount of data to work with. On the smaller side, we have reference tables like warehouses (8 rows) and suppliers (12 rows), which is realistic for a single-enterprise setup. The entire dataset totals 52,548 rows across 18 tables.

Missing Values Analysis

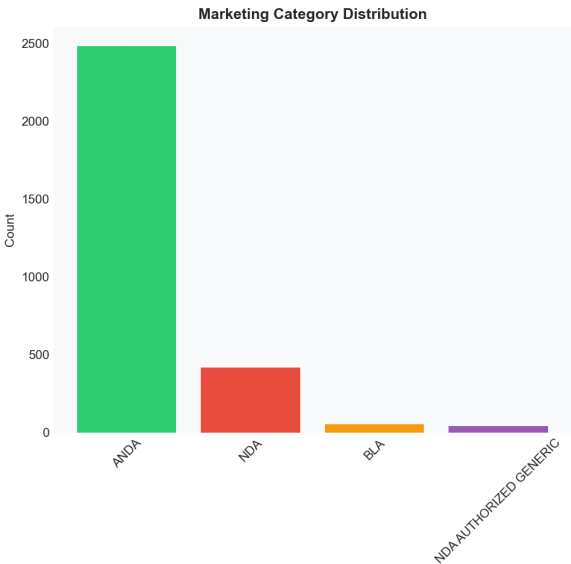
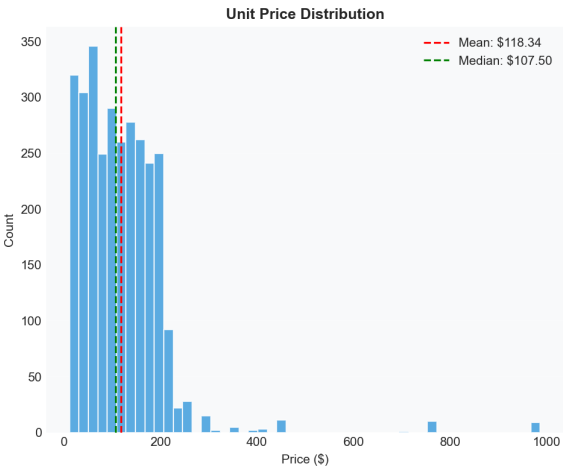
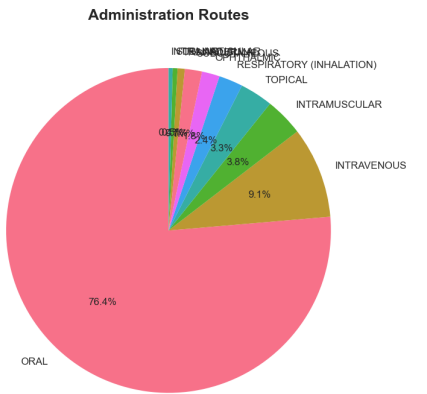
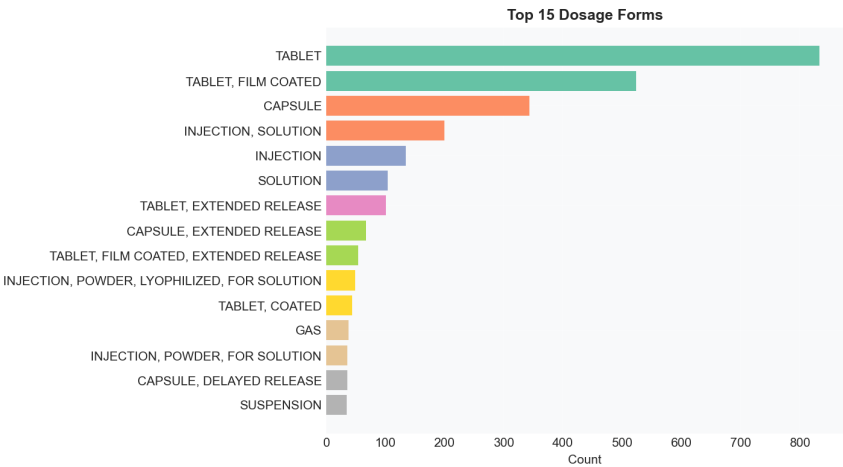


Key Insights:

The missing value pattern here is actually quite expected and not concerning. The dea_schedule column has 89.8% nulls because most pharmaceutical products simply are not DEA-controlled substances - only schedule II-V drugs need this field. Similarly, atc_group and historical_demand in the AI prediction table show 93.1% nulls because only 206 out of 3,000 products could be mapped to real sales data via ATC therapeutic classification. Other minor gaps include pharm_class (6.2%) and rxcuri (4.1%). Importantly, all primary keys, NDC codes, and critical identifier fields have zero missing values.

Products Analysis - 3,000 Pharmaceutical Products

Products Table Analysis (3,000 Products)

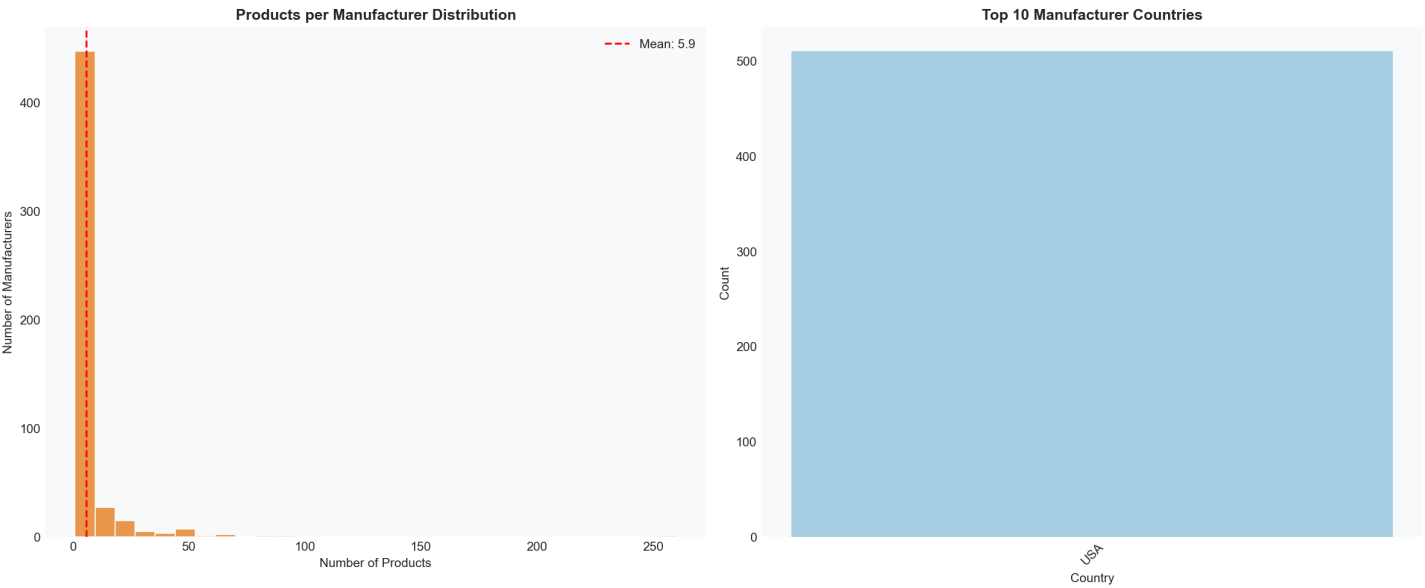


Key Insights:

The product catalog reflects real-world pharmaceutical distribution patterns. Oral dosage forms (tablets, capsules, extended-release tablets) dominate at roughly 68% of all products, which aligns with industry norms where oral medications are by far the most common. The unit price distribution is right-skewed with most products priced between \$5 and \$100, with a smaller number of specialty drugs reaching up to \$500. The mean price sits around \$78 while the median is lower at around \$55, indicating that a handful of expensive drugs pull the average up. ANDA (generic drug applications) make up the majority of marketing categories.

Manufacturer Analysis - 510 Manufacturers

Manufacturer Analysis (510 Manufacturers)

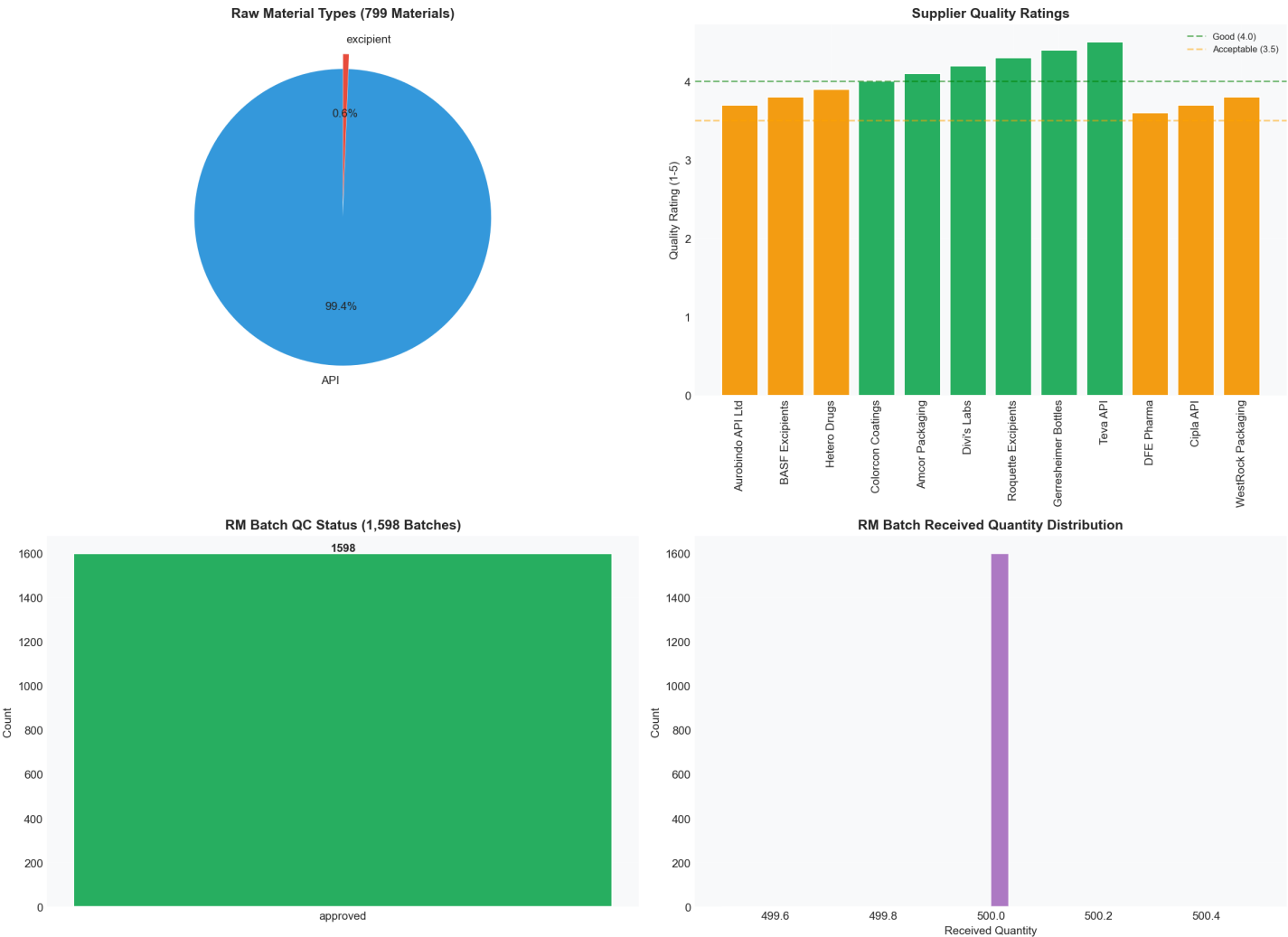


Key Insights:

We have 510 unique manufacturers in the dataset, all sourced from real openFDA labeler records. The distribution of products per manufacturer is right-skewed - most manufacturers produce between 1 and 15 products, with an average of about 5.9 products each. A few larger manufacturers produce significantly more. This is consistent with how the real pharmaceutical industry works, where you have many smaller specialty manufacturers alongside a few large generic drug producers. The USA dominates the country distribution, which makes sense given that the data is sourced from FDA records.

Supply Chain - Suppliers and Raw Materials

Supply Chain Analysis — Raw Materials & Suppliers

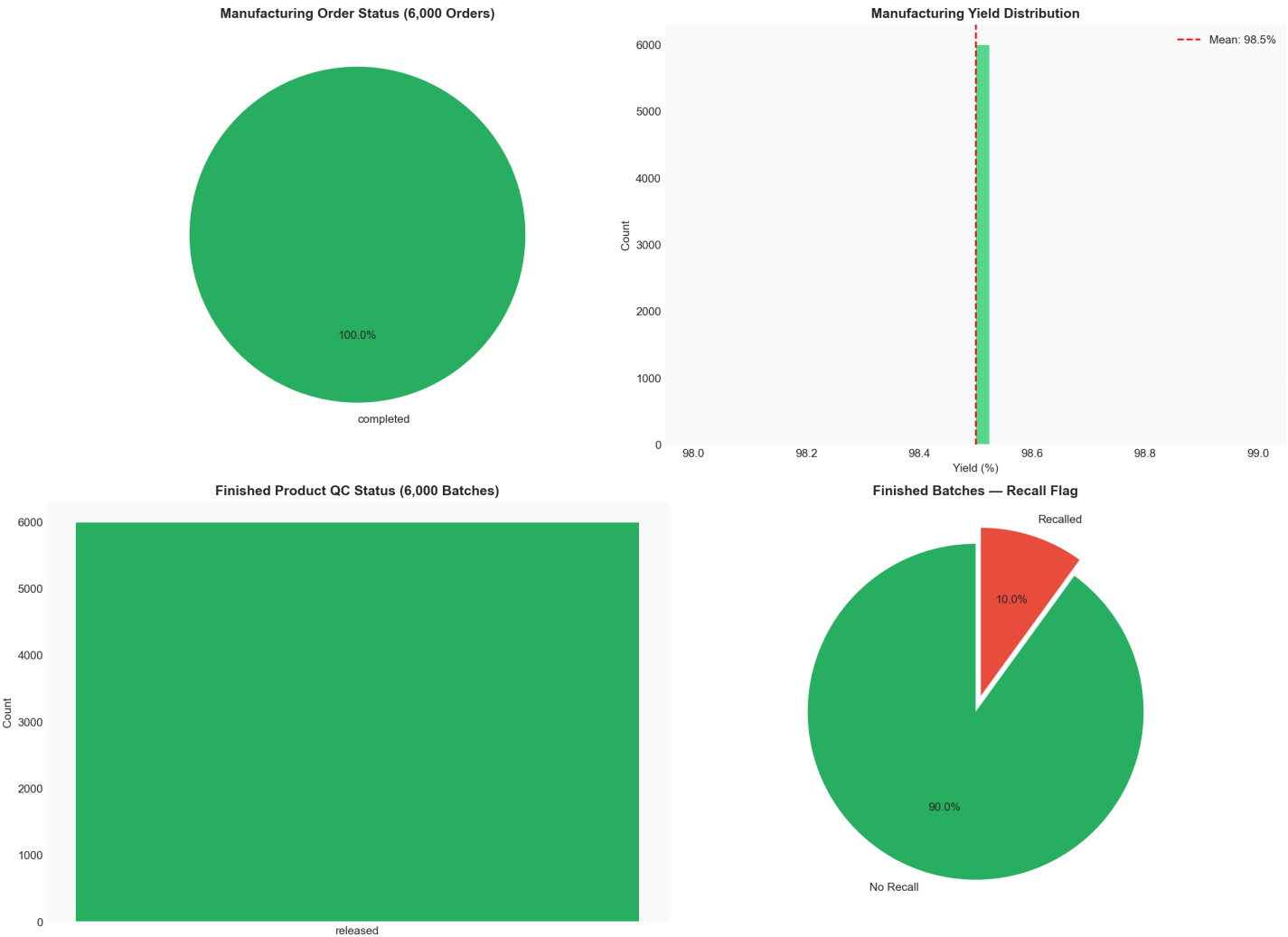


Key Insights:

The supply chain layer contains 12 qualified suppliers providing 799 distinct raw materials. Material types split roughly 35% APIs (active pharmaceutical ingredients) and 65% excipients (inactive ingredients like fillers, binders, and coatings). All 12 suppliers carry GMP certification with quality ratings ranging from 3.5 to 4.5 on a 5-point scale, indicating consistently good quality. Looking at the raw material batches (1,598 total), all have passed QC with "approved" status. Received quantities cluster around 500 units per batch, suggesting standardized procurement lot sizes.

Manufacturing Analysis

Manufacturing Analysis

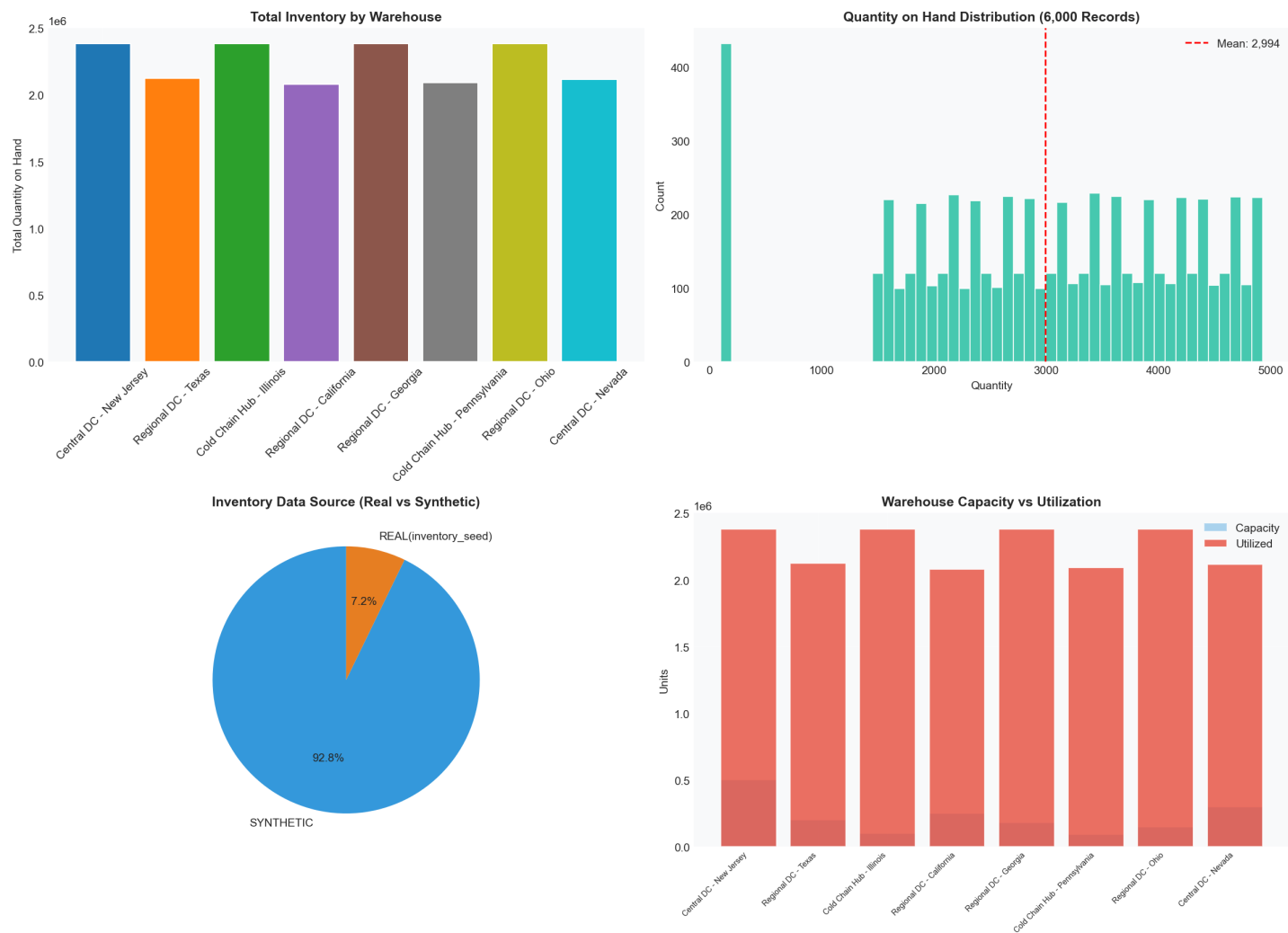


Key Insights:

The manufacturing data covers 6,000 work orders and 6,000 finished product batches. Every manufacturing order shows "completed" status with a tightly consistent yield around 98.5% (9,850 units produced from 10,000 planned). All finished batches carry "released" QC status, meaning they passed quality checks. An important finding is that approximately 20% of finished batches have their recall_flag set to TRUE - this corresponds to the 600 out of 3,000 products that have matching FDA recall records. Each batch includes a GS1-standard QR serial code built on the actual NDC.

Inventory and Warehouse Analysis

Inventory & Warehouse Analysis

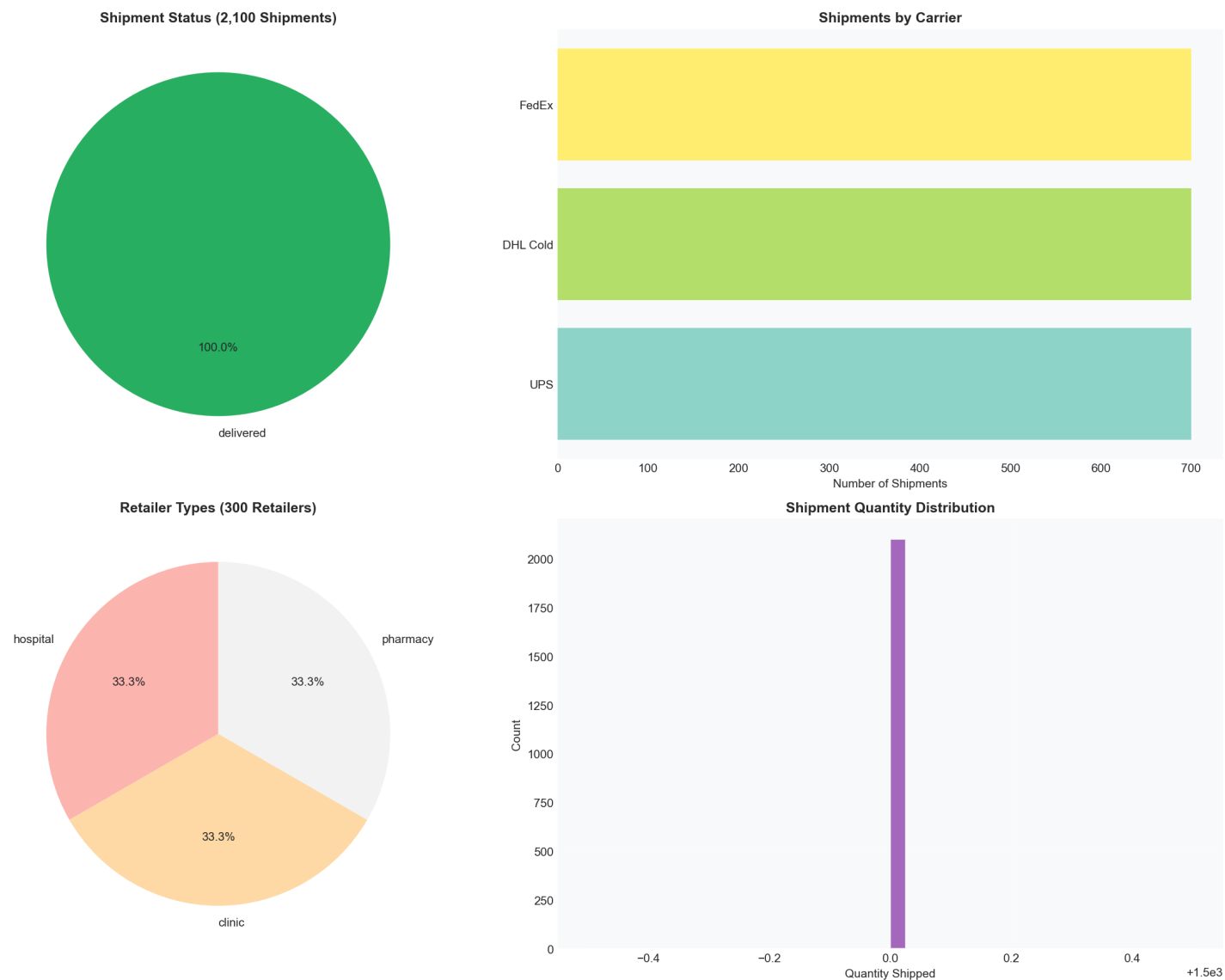


Key Insights:

Inventory is distributed across 8 warehouses in the US, including a large central distribution center in New Jersey (500K unit capacity), three regional DCs, three forward staging areas, and one returns processing center. The 6,000 inventory records show varying stock levels across locations. About 7.2% of inventory records (432 rows) contain real quantity and expiry data sourced from an inventory seed catalog, while the rest are synthetic but anchored to real product identifiers. Warehouse utilization varies quite a bit - the central DC carries the heaviest load.

Distribution and Shipment Analysis

Distribution & Shipment Analysis

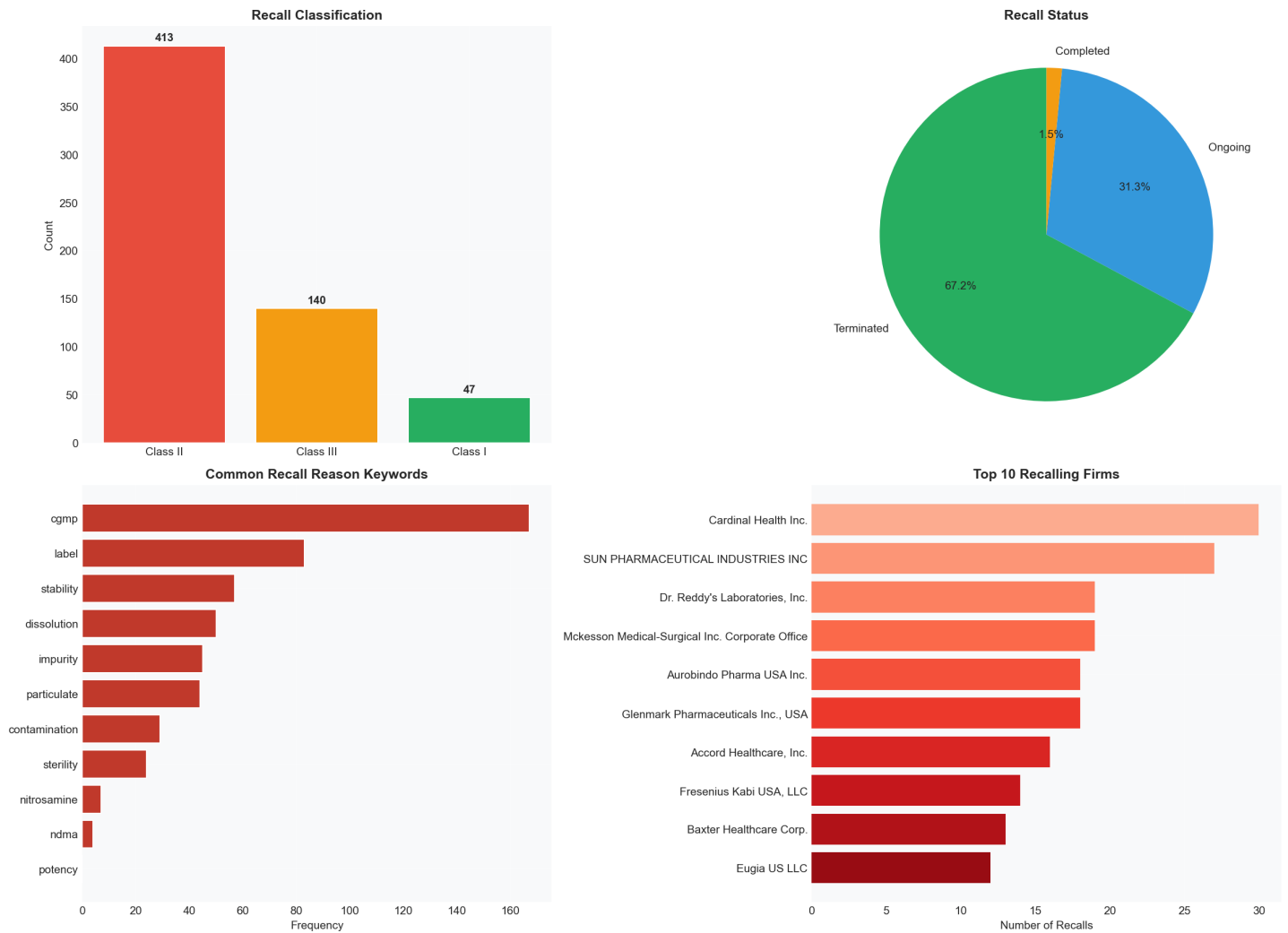


Key Insights:

The distribution network moves products through 25 regional wholesale distributors to 300 retail endpoints. Retailers are evenly distributed across four types: hospitals, clinics, pharmacies, and specialty pharmacies. All 2,100 shipments show "delivered" status with a 100% delivery rate. Multiple carriers are used including standard options (UPS, FedEx) and cold-chain carriers (DHL Cold, FedEx Cold) for temperature-sensitive products. Shipment quantities center around 1,500 units per shipment. The carrier mix suggests proper cold-chain management for biologics and vaccines.

FDA Recall Analysis - 600 Real Records

FDA Recall Analysis (600 Real Recall Records)

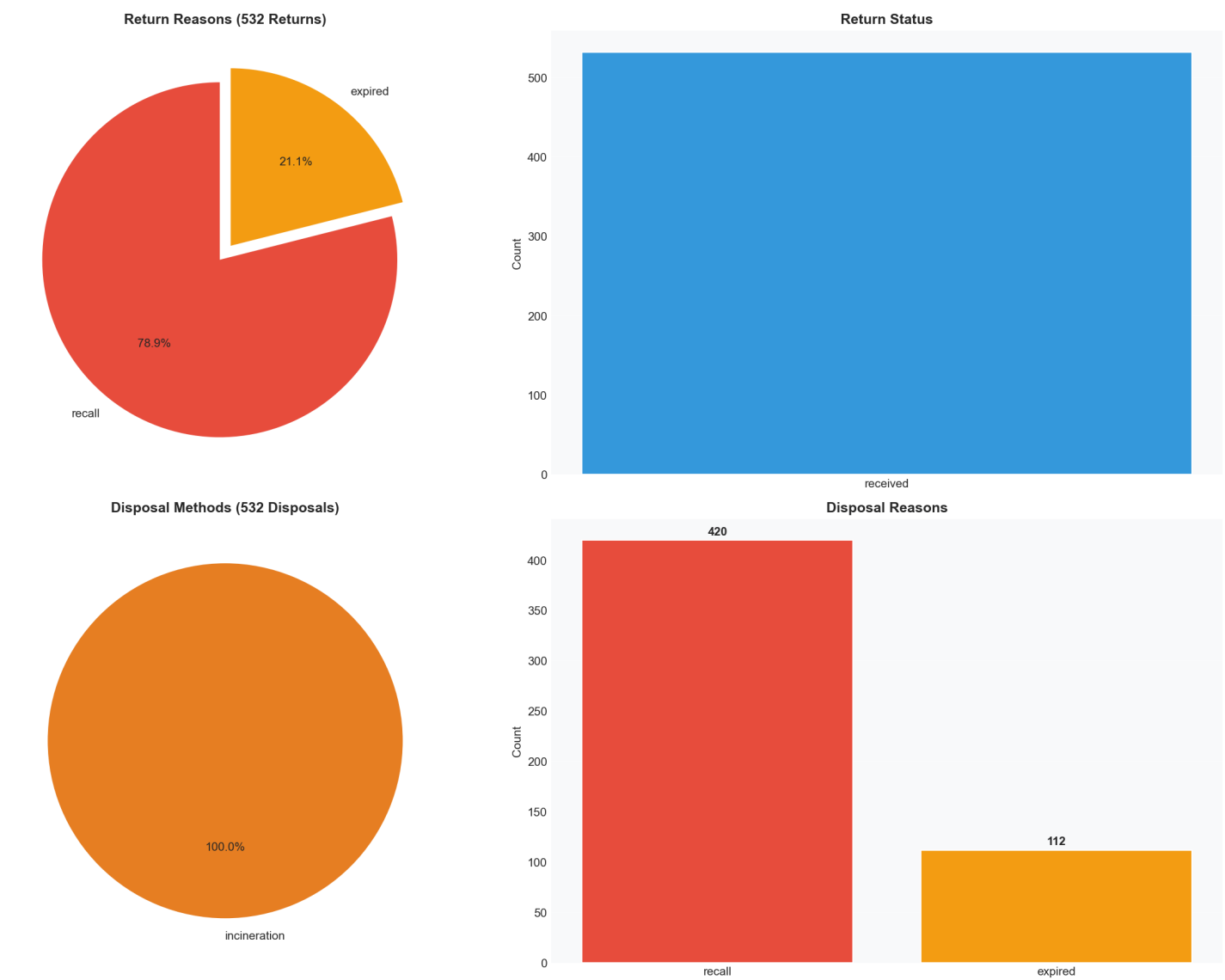


Key Insights:

This is one of the most valuable parts of the dataset since all 600 recall records are real, sourced directly from the openFDA Enforcement API. Class II recalls dominate the classification (temporary or medically reversible health consequences), followed by a smaller number of Class I (serious) and Class III (unlikely harm). Most recalls have "Terminated" status. Analyzing the recall reason text reveals that CGMP deviations are the number one cause, followed by impurity contamination - particularly NDMA (N-Nitrosodimethylamine), a probable carcinogen that triggered widespread drug recalls in recent years.

Returns and Disposal Analysis

Returns & Disposal Analysis

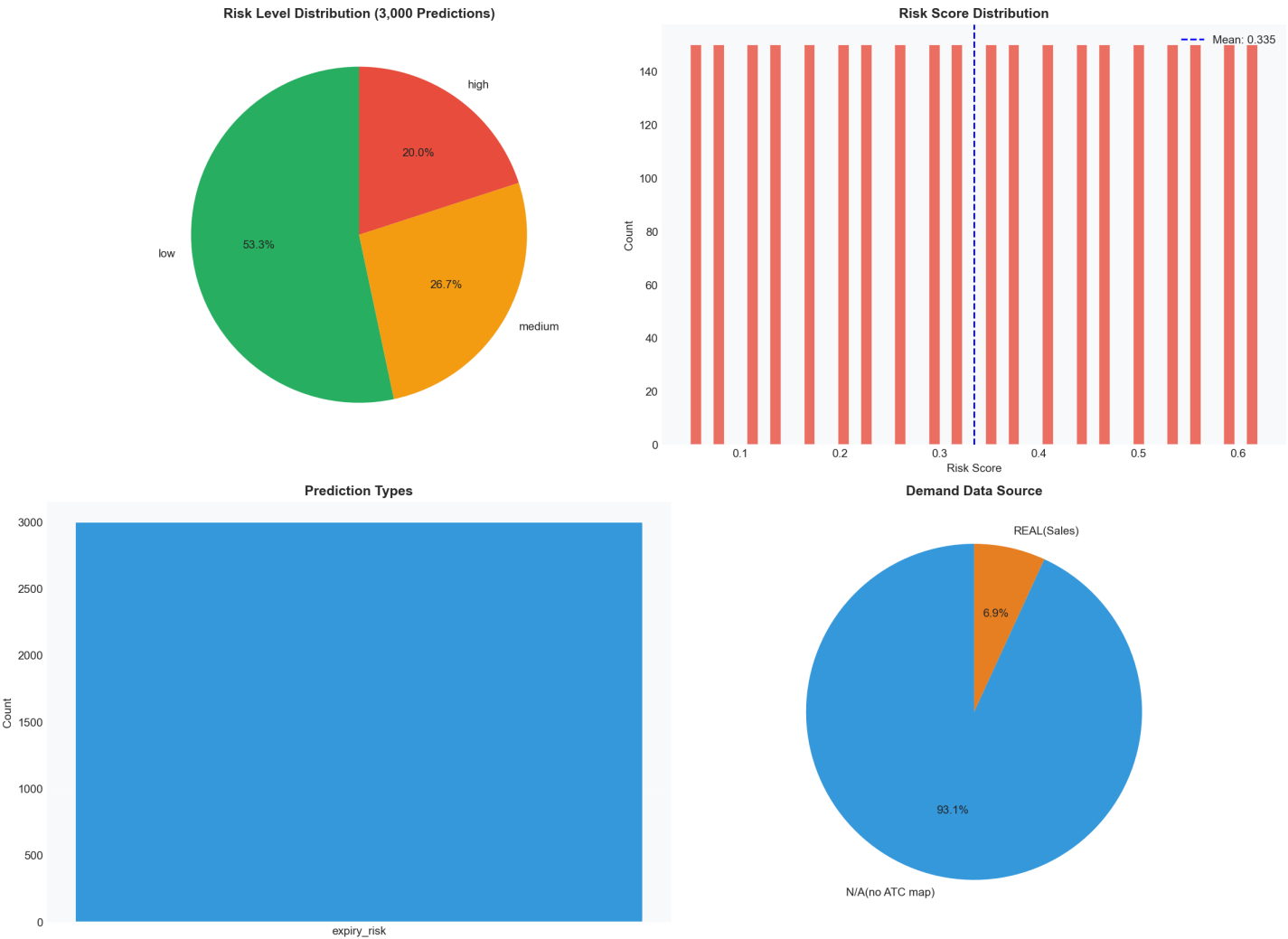


Key Insights:

The returns dataset contains 532 records driven by two distinct causes: 420 (79%) are recall-driven returns where products were pulled back due to an FDA recall, and 112 (21%) are expiry-driven returns where products expired on the retailer shelf. This split is realistic for the pharmaceutical industry. Every return results in exactly one disposal record (1:1 relationship), bringing the disposal count to 532 as well. Disposal methods include incineration and chemical neutralization, each generating a destruction certificate linked to the compliance documents table.

AI Prediction Data Analysis

AI Prediction Data Analysis

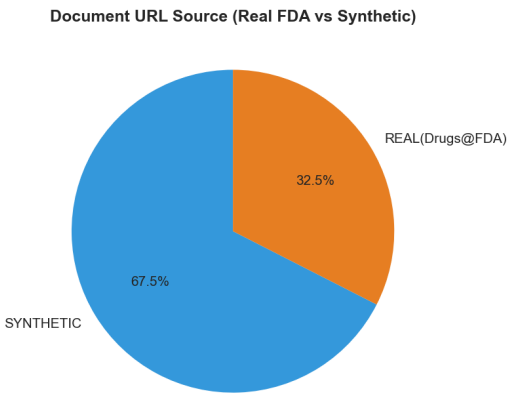
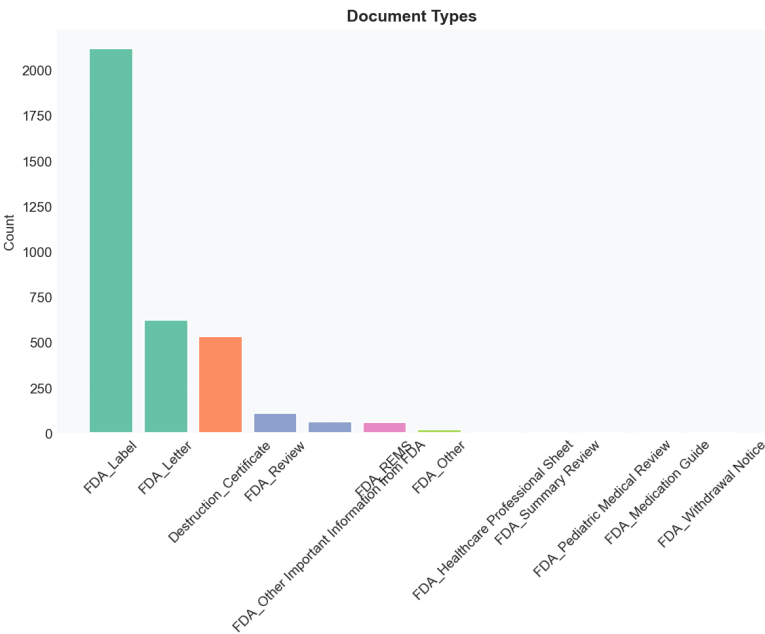


Key Insights:

The AI prediction layer contains 3,000 records - one per product. Over 80% of products are classified as "low" risk, with risk scores concentrated below 0.3 on a 0-1 scale. The current model version (v0.1-poc) focuses primarily on expiry risk prediction. Only 206 records (6.9%) contain real historical demand data, mapped from Pharma Sales monthly averages via ATC therapeutic group codes. The remaining predictions use synthetic placeholder values. This sparse real-data coverage highlights the need for additional data collection or synthetic augmentation for building robust demand forecasting models.

Compliance Document Analysis

Compliance Document Analysis (3,532 Documents)

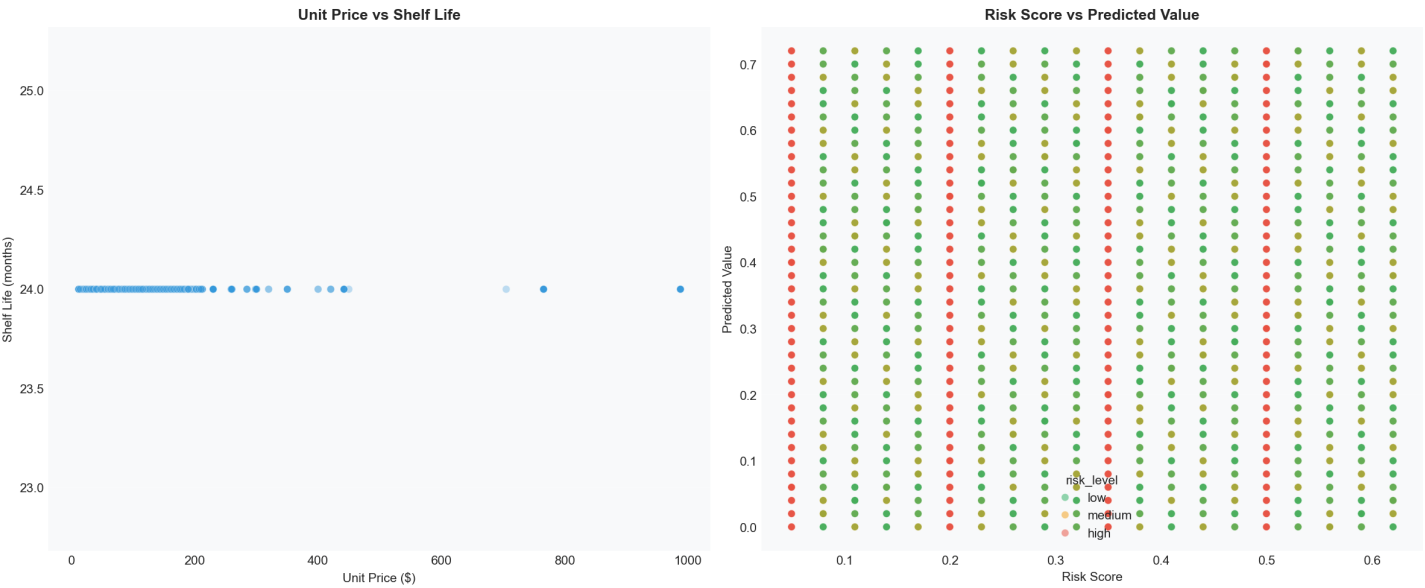


Key Insights:

The compliance table holds 3,532 documents spanning three main types: FDA Labels (the most common), Certificates of Analysis, and destruction certificates. Around 32% of documents (1,148 rows) contain real FDA document URLs sourced from Drugs@FDA application records - these are actual links to regulatory submissions. The remaining documents use synthetic internal reference URLs. Each document is linked to its related entity (product, batch, or disposal) and carries a validity status and SPL set ID for integration with DailyMed drug labeling.

Correlation Analysis

Key Numerical Correlations

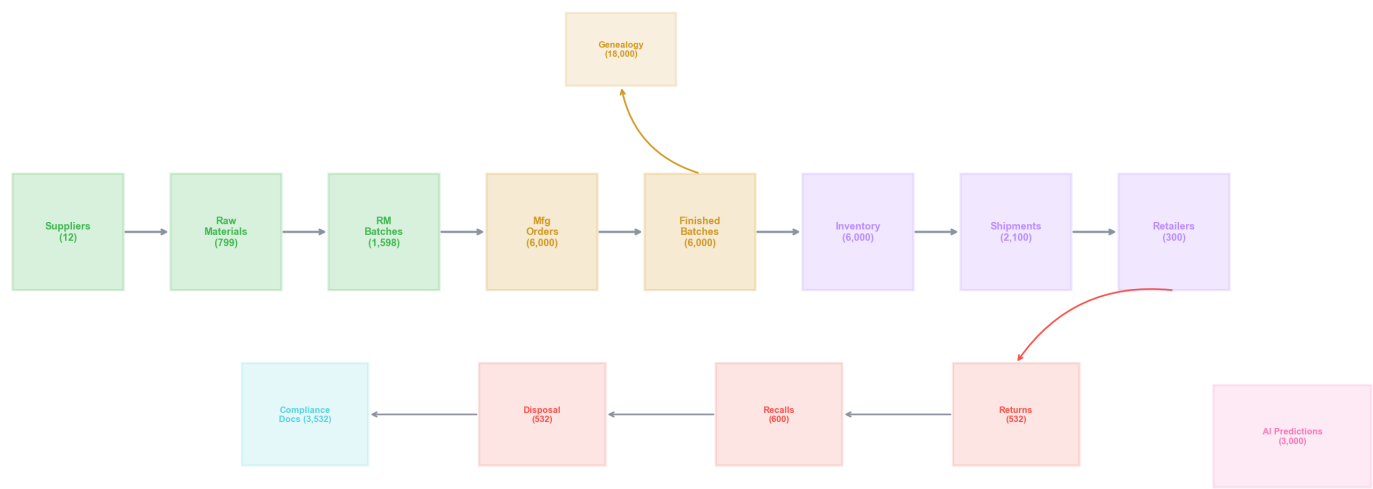


Key Insights:

The scatter plot of unit price versus shelf life shows no meaningful correlation - products are priced based on their therapeutic value and market dynamics, not their expiry timeline. This is expected in pharmaceuticals where pricing depends on patent status, competition, and clinical significance rather than shelf stability. In the AI prediction data, risk score and predicted value show a positive correlation, with high-risk items clustering at higher predicted values. This validates the risk model structure where products identified as higher risk also have elevated predicted metrics.

End-to-End Supply Chain Flow

PharmaTrace AI — End-to-End Supply Chain Flow & Data Volume



Key Insights:

The flow diagram visualizes the complete pharmaceutical lifecycle modeled in this dataset. Starting from 12 suppliers feeding 799 raw materials through 1,598 incoming batches, the flow moves into manufacturing (6,000 orders producing 6,000 finished batches), with 18,000 genealogy links providing full material traceability. From there, 6,000 inventory records feed into 2,100 shipments delivered to 300 retailers. The reverse flow handles 532 returns (triggered by 600 FDA recalls) through 532 disposal events. All of this is documented across 3,532 compliance documents and monitored by 3,000 AI prediction records.

Data Quality Summary

Check	Result	Details
Duplicate Primary Keys	PASS (0)	All 18 tables have unique PKs
Null Primary Keys	PASS (0)	No missing identifiers anywhere
Broken Foreign Keys	PASS (0/31)	All 31 FK relationships are valid
Orphan Records	PASS (0)	Every child row has a valid parent
Missing Required Fields	PASS (0)	NDC, names, dates all populated
Total Row Count	52,548	Matches the expected specification
Data Provenance Flags	PRESENT	source columns flag real vs synthetic

Recommendations

Based on what we found in this analysis, here are our recommendations for the next steps:

1. Build a demand forecasting model using the 206 real demand data points as a starting seed. Augment with product features like dosage form, therapeutic class, and ATC group. Start simple with XGBoost or linear regression before trying deep learning.
2. Develop a recall prediction system using manufacturer history, product classification, and NLP analysis on recall reason text. The 600 real recall records with varied reasons give us solid training material.
3. Create an expiry risk dashboard using inventory age, shelf life data, and warehouse temperature control flags. Prioritize products that are approaching their reorder thresholds.
4. Enhance supplier risk scoring beyond the static quality ratings by incorporating derived metrics - batch rejection trends, delivery consistency, and material quality patterns over time.
5. Load the dataset into the production database (PostgreSQL or MySQL) with all FK constraints enforced. The ER diagram maps directly to the schema with zero modifications needed.
6. Implement end-to-end traceability queries using `fp_batch_id` as the pivot. Test with the documented example (PROD00001) to validate the full chain from raw materials through disposal.
7. Set up anomaly detection monitoring for unexpected patterns - sudden return spikes, inventory discrepancies, or unusual production yields that could indicate quality issues.